Syllabus-Matched Questions List

Filtered from questions.pdf based on Course Topics (Topics 01 – 12)

> Summary of Match:
> • Out of 188 questions in questions.pdf, a total of 97 questions are covered within your 12 course topic PDFs.
> • Out-of-syllabus questions (IP Subnetting calculations, Transport/Application Layer deep dives like TCP/UDP internals, DNS, Email, Firewalls, Cryptography, IPSec) have been excluded.

Topic 01: Introduction & Basic Concepts

Q1	What is a computer network?
Q2	Discuss various types of networks topologies in computer network.
Q3	Also, discuss advantages and disadvantages of each topology.
Q4	What are the applications of Computer Networks?
Q5	What is OSI Model? Explain the functions, protocols and services of each layer?
Q6	Explain the following: I. LAN, II. MAN, III. WAN
Q7	What are the network components? What do you mean by client-server network?
Q8	Write some protocol names of each layer of TCP/IP model.
Q9	Explain protocol, layer, peer and interface using layered diagram.
Q10	Explain connection multiplexing, connectionless and connection-oriented services.
Q11	Explain OSI and TCP/IP model
Q12	Describe HDLC frame format.
Q13	How many types of HDLC frame there are? Explain control field formation of each of those.

Topic 02: Point-to-Point Protocol & Medium Access Control

Q14	When and why we use point-to-point connection?
Q15	Draw the PPP frame format. Explain each field.
Q16	Explain the diagram for bringing a line up and down in a PPP communication.
Q17	ATM has how many cell formats? Explain the commonly used one.
Q18	Why Standardization is necessary? Explain with example
Q19	How does CSMA/CD work? Explain

Q20	What do you mean by token? Explain IEEE 802.4 and 802.5.

Topic 03: LAN Interconnection & Layer 2 Devices

Q21	Why not just one big LAN?
Q22	What devices are available for LAN's interconnection? Mention which layer they work.
Q23	Explain HUB and Bridge. Compare them.
Q24	Give a bridge learning example.
Q25	What will happen if there is a loop between two nodes?
Q26	What is spanning tree? How it helps bridge.
Q27	Compare and contrast between routers and bridge.
Q28	What do you mean by cut-through switching?
Q29	Compare and contrast between switch and bridge.
Q30	What is FDDI? How does it work?
Q31	What are the advantages of FDDI?
Q32	Draw the FDDI frame formation.
Q33	Differentiate between Ethernet and token ring.

Topic 04: Network Layer: Routing Algorithms & Congestion Control

Q35	What are the physical media available for Ethernet, Fast Ethernet and Gigabit Ethernet? Also note their maximum segment length.
Q36	What leads to Fast Ethernet?
Q37	What are the Ethernet, Fast Ethernet and Gigabit physical media types? Write down their maximum-segment-length.
Q38	Explain Non-adaptive & adaptive routing algorithm.
Q39	What do you mean by the optimality principle and sink tree?
Q40	What do you mean by routing metrics? Explain with figure and example.
Q41	Use shortest path algorithm to find shortest path from A to M.
Q42	State the steps of the shortest path algorithm.
Q43	Explain with figure the count to infinity problem in DVR algorithm.
Q44	Briefly explain the steps of link state routing.
Q45	Why we need interconnecting networks and how to do that? How the available networks differ from each other?

Q46	Why we need packet fragmentation? Do fragmentation for the given network (SITU-3000 to MTU-1000/4000).
Q47	What are the fragmentation related fields? Explain.

Topic 05: Network Layer Protocols (ARP, DHCP, IPv4/v6, Mobile IP)

Q48	What are the network layer protocols? What are purposes of using an IP address?
Q49	Why deliveries of a packet need two levels of addressing e.g. IP and MAC?
Q50	Explain static and dynamic mapping. State their limitations.
Q51	Show with figure that ARP request is broadcast and reply is unicast.
Q52	Show ARP packet formation and encapsulation. What are the 4 cases of using ARP.
Q53	A host with IP address 130.23.3.20 and physical address B23455102210 has a packet to send to another host with IP address 130.23.43.25 and physical address A46EF45983AB. The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames.
Q54	Show RARP packet formation and encapsulation.
Q55	Explain using figure the four cases of using ARP.
Q56	What is the purpose of DHCP protocol and how does it work?
Q57	Draw state-transition diagram for DHCP. Explain all six states.
Q58	Show the DHCP message exchange between client-server.
Q59	Draw IPv4 and IPv6 datagram format and explain about its each field briefly.
Q60	Define MTU and the fields of fragmentation with appropriate figure.
Q61	What are the merits of IPv6 over IPv4?
Q62	Explain three transition strategies from IPv4 to IPv6.
Q63	Explain the fragmentation procedure of IPv6 packet and compare it to IPv4 fragmentation.
Q64	Explain Home address and care-of address, Home agent and foreign agent with appropriate figure.
Q65	Explain multiplexing and de-multiplexing between network and transport layer using appropriate figure.
Q66	To communicate with a remote host, a mobile host goes through three phases: agent discovery, registration, and data transfer. Time diagram.
Q67	Communication involving mobile IP can be inefficient. Explain the scenarios.

Topic 06: Wireless LAN & Network Devices

Q68	Differentiate between ATM cell and TCP packet / Wireless & Wired LAN components.

Topic 07 & 08: WAN Technologies (ISDN, X.25, Frame Relay, DSL, Cable, Satellite)

Q69	Describe both the ATM cell types with appropriate cell formation figure.

Topic 09: Mobile Telephone Systems & High-Speed Ethernet

Q112	Differentiate between 1G, 2G, 3G and 4G.
Q113	Write short notes on IMPS and AMPS.
Q114	How AMPS accumulate more calls than IMPS?
Q115	Step by step explain MOC with flow diagram.
Q116	Draw the basic diagram of GSM system. Show the channel distribution and framing structure of GSM network.
Q117	What are the improvements in IMT-2000 network?
Q118	How to handle a call during cell change. Explain soft/hard handoff with figure.
Q120	What are 3 choices of Fast-Ethernet?
Q121	Write short notes on: 100 BASE T4, 100 Base TX, 100 Base FX
Q122	Write short notes on: a. 1000 BASE SX, b. 1000 BASE LX, c. 1000 BASE CX, d. 1000 BASE T, e. 8B/10B Encoder, f. A buffered distributor
Q123	Explain with figure the task of MII and GMII.

Topic 10: Routing Protocols, Interworking & VLANs

Q107	How to measure the network performance.

Topic 11: NAT, SNMP, and VPN

Q142	Write short note on SNMP, MIBs, TRAP. What is the purpose of MIBs?
Q143	Show SNMP manager and an SNMP agent communicate using the SNMP protocol, also explain the meaning of each message

Topic 12: Enterprise Network Implementation

Q111	Explain the characteristics of enterprise network and design issues for this network.